CS 352 @ YZU
資料庫系統概論

2025/01/08

DBMS PROPOSAL

LinkUp：交友配對與活動揪團平台

第三組

OUTLINE

- Introduction
- System Architecture
- Database Description
- Demonstration
- Others
  - Future Work

Introduction

Problems & Target Audiences



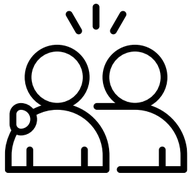
想要單純交友？

現有的交友軟體越來越「不單純」，介面也過於花俏、太多「照騙」



想要找人出遊？

某些行程或活動想找人一起去，但朋友沒空或是沒興趣



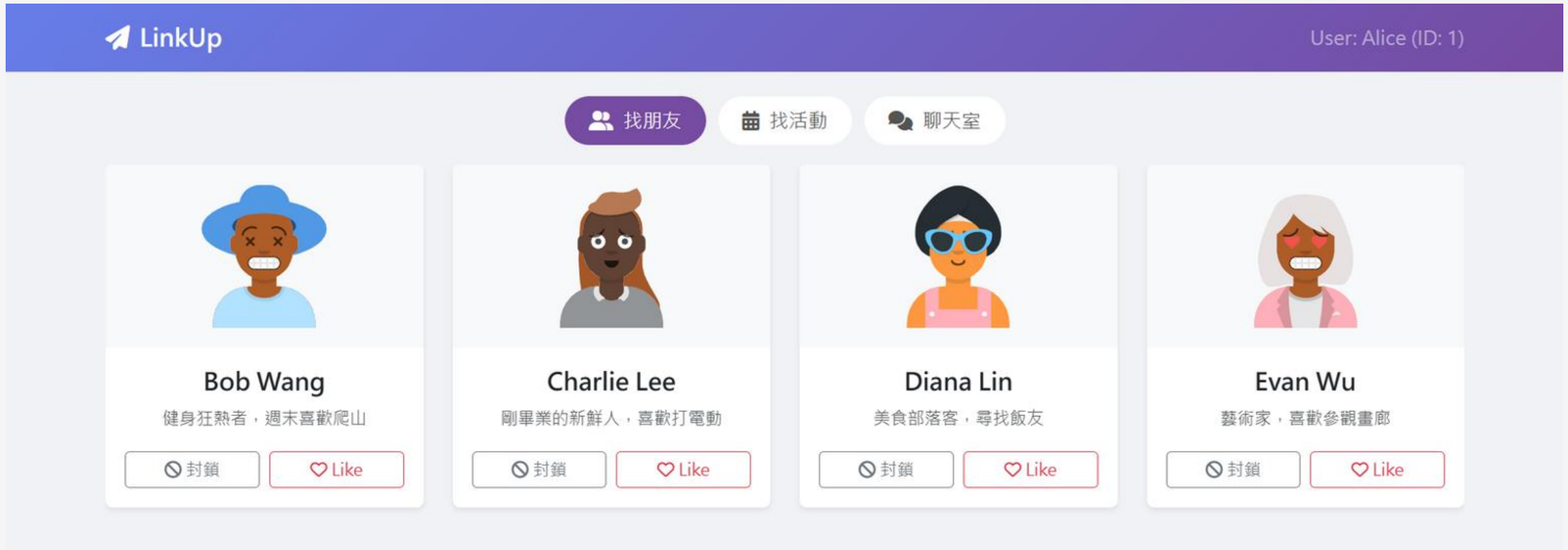
想找人聊聊天？

有些事情不想跟認識的人分享，或單純想和陌生人聊天、認識新朋友

What is "LinkUp"?

交友配對與活動揪團平台

- **找朋友**：可以查看所有用戶，從個人簡介與照片中找尋想認識的對象



What is "LinkUp"?

交友配對與活動揪團平台

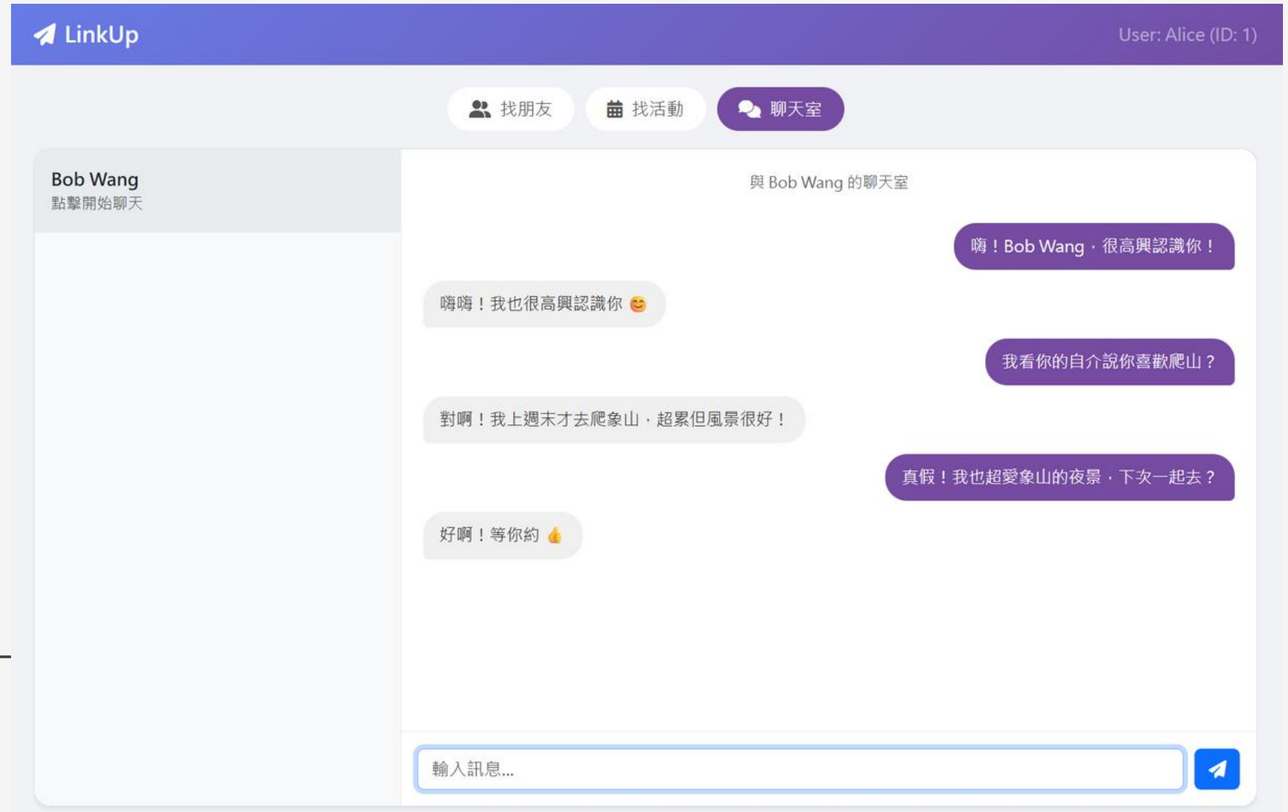
- **找活動**：用戶可以自由參加其他使用者發起的活動



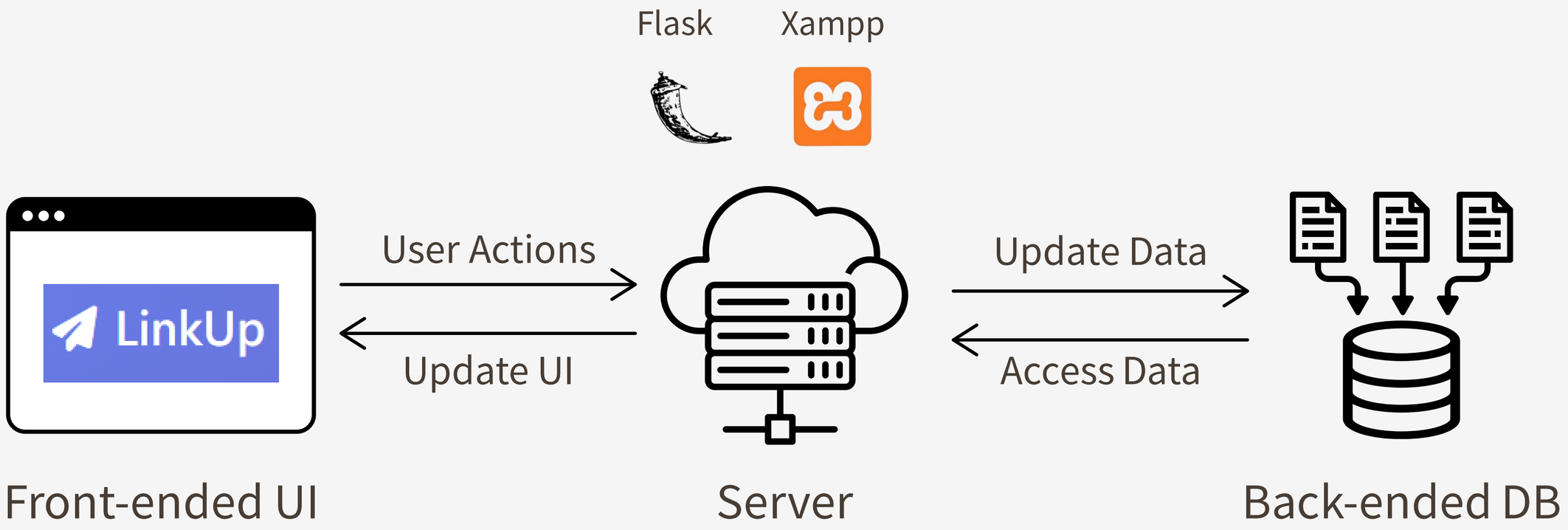
What is "LinkUp"?

交友配對與活動揪團平台

- **聊天室**：兩位用戶互相感興趣後，就能以朋友身份私訊聊天

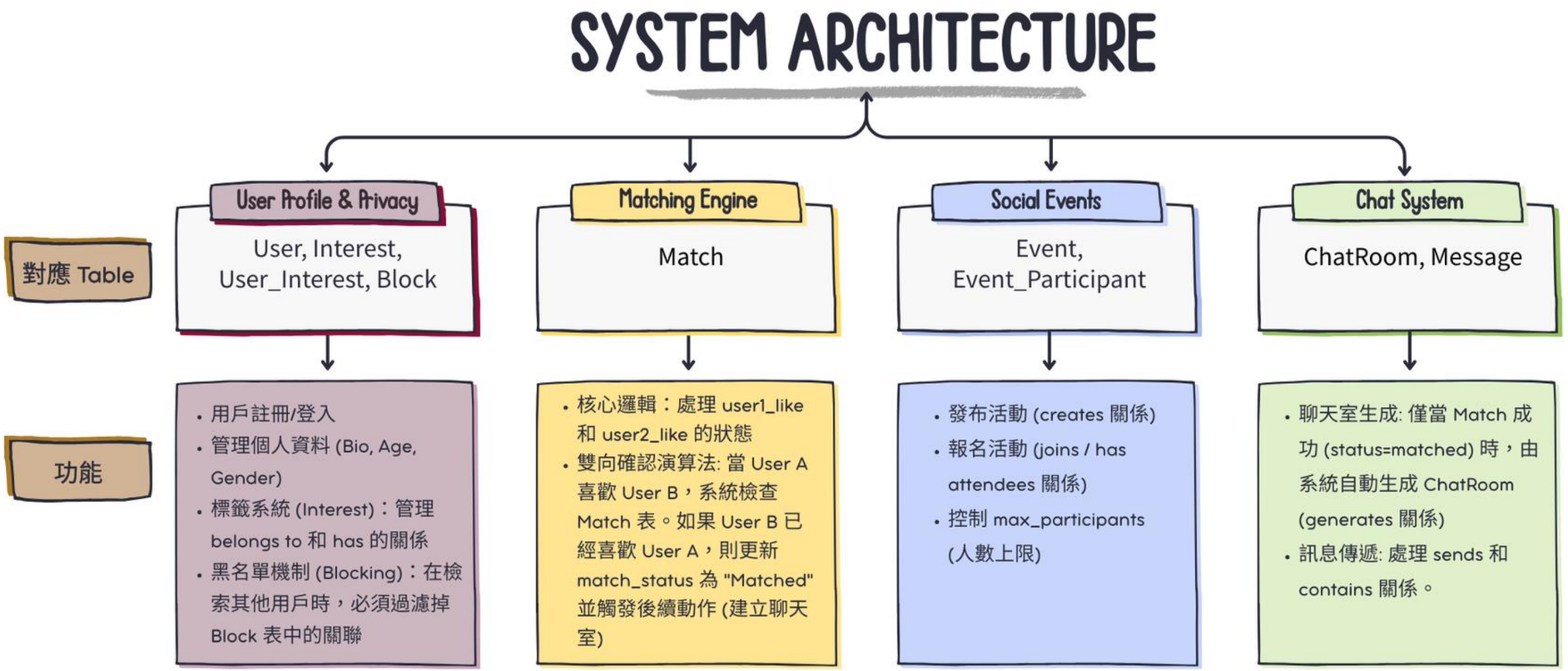


How "LinkUp" Works?



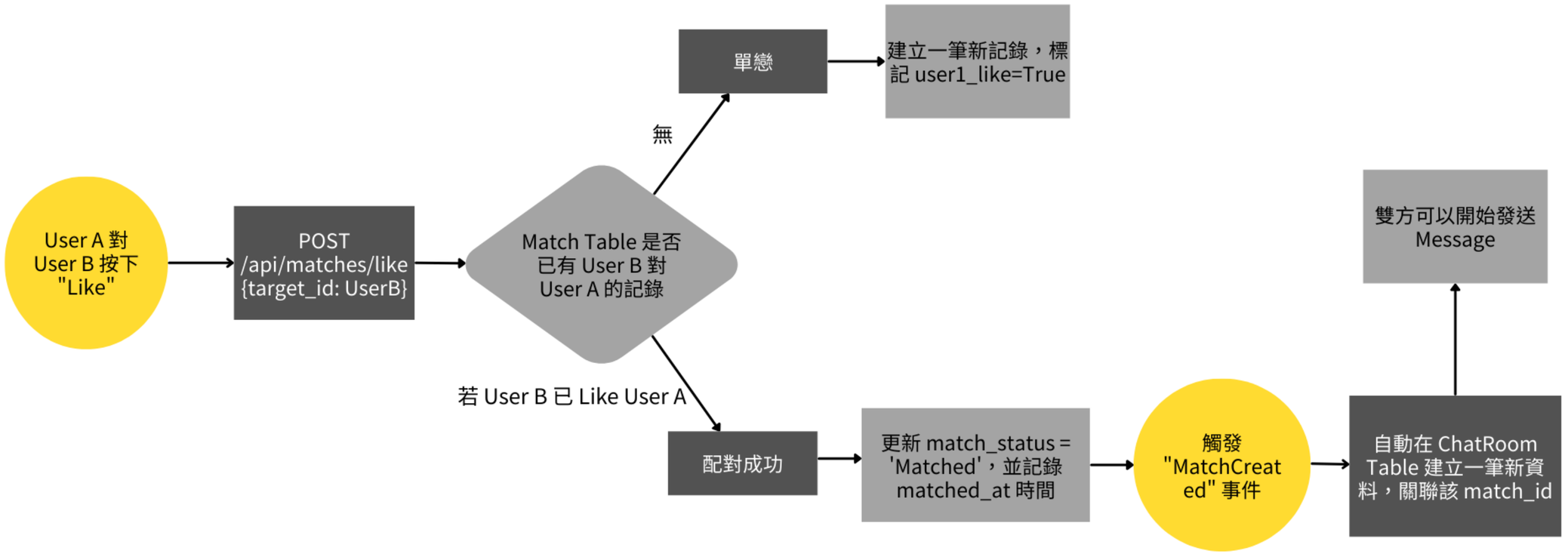
System Architecture

Overview


SYSTEM ARCHITECTURE
對應 Table
功能
User Profile & Privacy
User, Interest, User_Interest, Block
Matching Engine
Match
Social Events
Event, Event_Participant
Chat System
ChatRoom, Message
• 用戶註冊/登入
• 管理個人資料 (Bio, Age, Gender)
• 標籤系統 (Interest)：管理 belongs to 和 has 的關係
• 黑名單機制 (Blocking)：在檢索其他用戶時，必須過濾掉 Block 表中的關聯
• 核心邏輯：處理 user1_like 和 user2_like 的狀態
• 雙向確認演算法: 當 User A 喜歡 User B，系統檢查 Match 表。如果 User B 已經喜歡 User A，則更新 match_status 為 "Matched" 並觸發後續動作 (建立聊天室)
• 發布活動 (creates 關係)
• 報名活動 (joins / has attendees 關係)
• 控制 max_participants (人數上限)
• 聊天室生成: 僅當 Match 成功 (status=matched) 時，由系統自動生成 ChatRoom (generates 關係)
• 訊息傳遞: 處理 sends 和 contains 關係。

Match & Chat Flow

配對成功與聊天室建立
(The "Match & Chat" Flow)



Event Participation Flow

活動報名
(Event Participation Flow)



Database Description

Design Phase

- User
- User_Interest
- Interest
- Block

以單一用戶為出發點，設計用戶相關表格

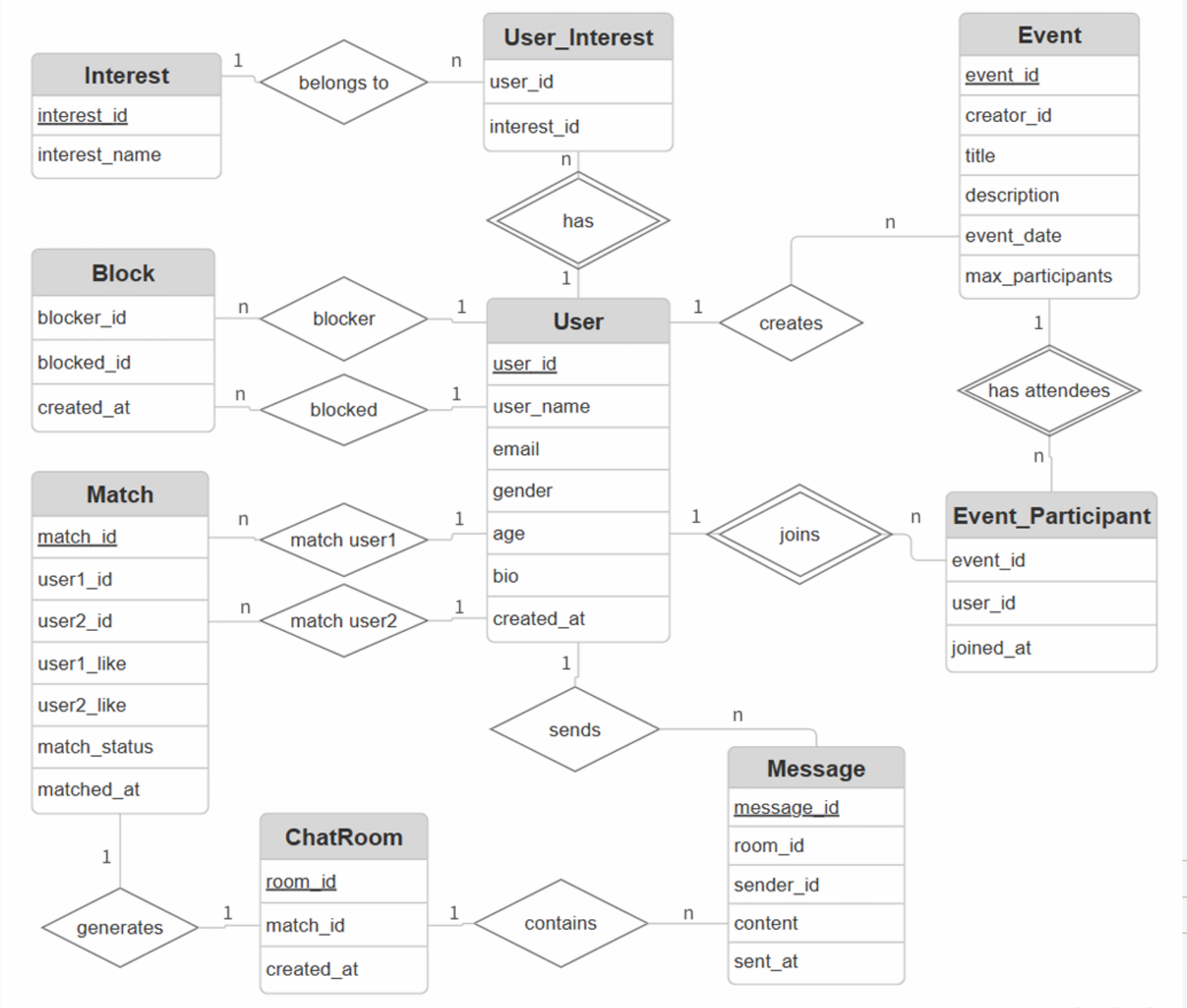
- Match
- ChatRoom
- Message

兩個用戶可以結為朋友、互相聊天

- Event
- Event_Participant

兩個或以上用戶可以揪團參加活動

ER Model


Interest
interest_id
interest_name
belongs to
User_Interest
user_id
interest_id
has
Event
event_id
creator_id
title
description
event_date
max_participants
Block
blocker_id
blocked_id
created_at
blocker
blocked
User
user_id
user_name
email
gender
age
bio
created_at
creates
has attendees
Match
match_id
user1_id
user2_id
user1_like
user2_like
match_status
matched_at
match user1
match user2
joins
Event_Participant
event_id
user_id
joined_at
sends
Message
message_id
room_id
sender_id
content
sent_at
generates
ChatRoom
room_id
match_id
created_at
contains
1
n

Relation Model

- Relation Model
  - User(user_id, user_name, email, gender, age, bio, created_at)
  - User_Interest(user_id, interest_id)
  - Interest(interest_id, interest_name)
  - Block(blocker_id, bolcked_id, created_at)
  - Match(match_id, user1_id, user2_id, user1_like, user2_like, match_status, matched_at)
  - ChatRoom(room_id, match_id, created_at)
  - Message(message_id, room_id, sender_id, content, sent_at)
  - Event(event_id, creator_id, title, description, event_date, max_participants)
  - Event_Participant(event_id, user_id, joined_at)

Relation Model (Cont.)

- Primary Keys
  - PK(User) = user_id
  - PK(User_Interest) = user_id, interest_id
  - PK(Interest) = interest_id
  - PK(Block) = blocker_id, blocked_id
  - PK(Match) = match_id
  - PK(Chat_Room) = room_id
  - PK(Message) = message_id
  - PK(Event) = event_id
  - PK(Event_Participant) = event_id, user_id

Relation Model (Cont.)

- Foreign Keys
  - User_Interest.user_id → User.user_id
  - User_Interest.interest_id → Interest.interest_id
  - Block.blocker_id → User.user_id
  - Block.blocked_id → User.user_id
  - Event_Participant.event_id → Event.event_id
  - Event_Participant.user_id → User.user_id

Demonstration

Users

	user_id	user_name	email	gender	age	bio	created_at
Edit Copy Delete	1	Alice Chen	alice@example.com	Female	25	喜愛攝影與旅行的工程師	2026-01-01 10:00:00
Edit Copy Delete	2	Bob Wang	bob@example.com	Male	28	健身狂熱者，週末喜歡爬山	2026-01-02 11:30:00
Edit Copy Delete	3	Charlie Lee	charlie@example.com	Male	24	剛畢業的新鮮人，喜歡打電動	2026-01-03 09:15:00
Edit Copy Delete	4	Diana Lin	diana@example.com	Female	27	美食部落客，尋找飯友	2026-01-04 14:20:00
Edit Copy Delete	5	Evan Wu	evan@example.com	Non-binary	26	藝術家，喜歡參觀畫廊	2026-01-05 16:45:00

(User Table)

	user_id	interest_id
Edit Copy Delete	1	1
Edit Copy Delete	1	2
Edit Copy Delete	2	2
Edit Copy Delete	3	3
Edit Copy Delete	4	4
Edit Copy Delete	5	5

(User_Interest Table)

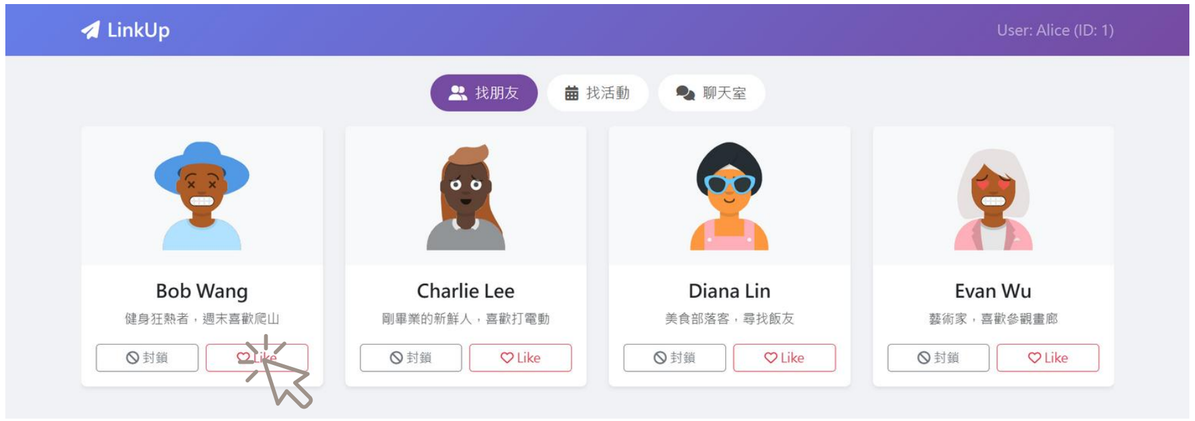
	interest_id	interest_name
Edit Copy Delete	5	Art
Edit Copy Delete	4	Foodie
Edit Copy Delete	3	Gaming
Edit Copy Delete	2	Hiking
Edit Copy Delete	1	Photography

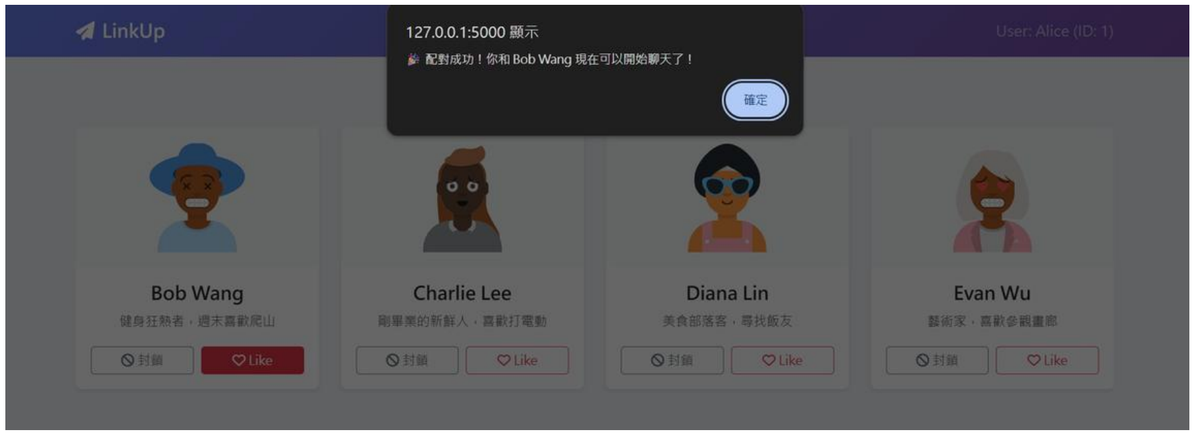
(Interest Table)

Match

	match_id	user1_id	user2_id	user1_like	user2_like	match_status	matched_at
Edit Copy Delete	1	1	2	1	1	matched	2026-01-05 15:50:15

(Match Table)





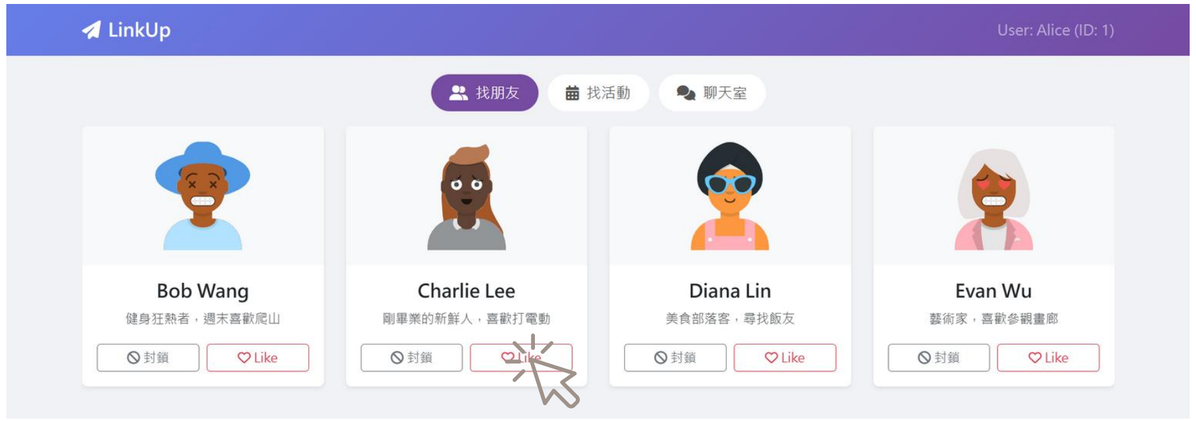
	room_id	match_id	created_at
Edit Copy Delete	1	1	2026-01-05 15:50:15

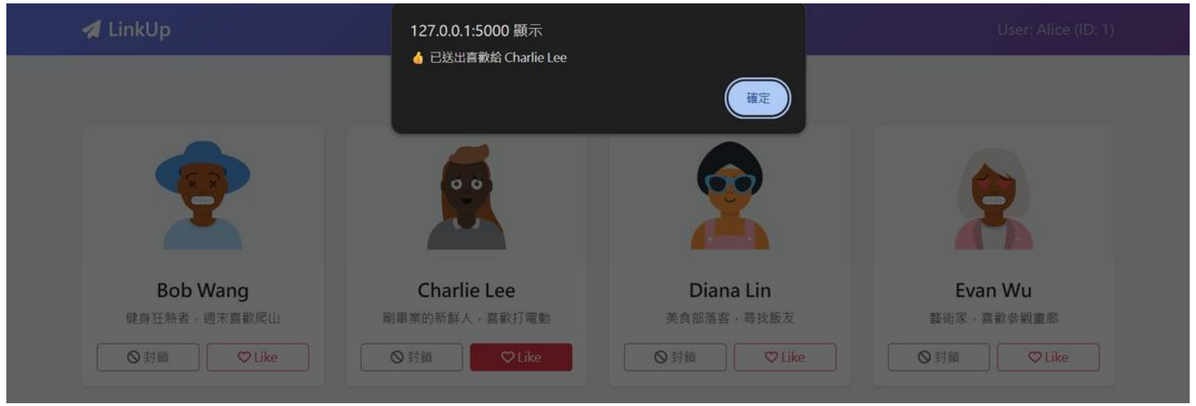
(ChatRoom Table)

Like

←T→	match_id	user1_id	user2_id	user1_like	user2_like	match_status	matched_at
Edit Copy Delete	1	1	2	1	1	matched	2026-01-05 15:50:15
Edit Copy Delete	2	1	3	1	0	pending	*NULL*

(Match Table)

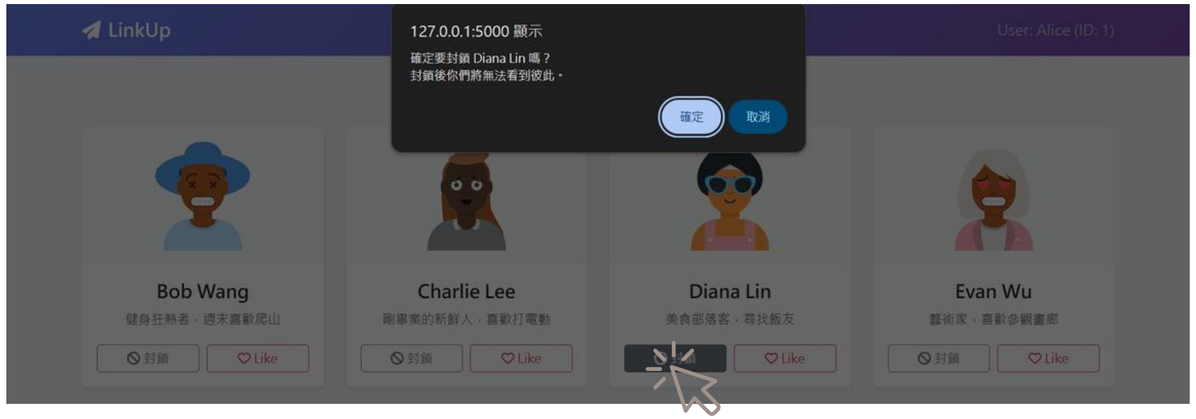


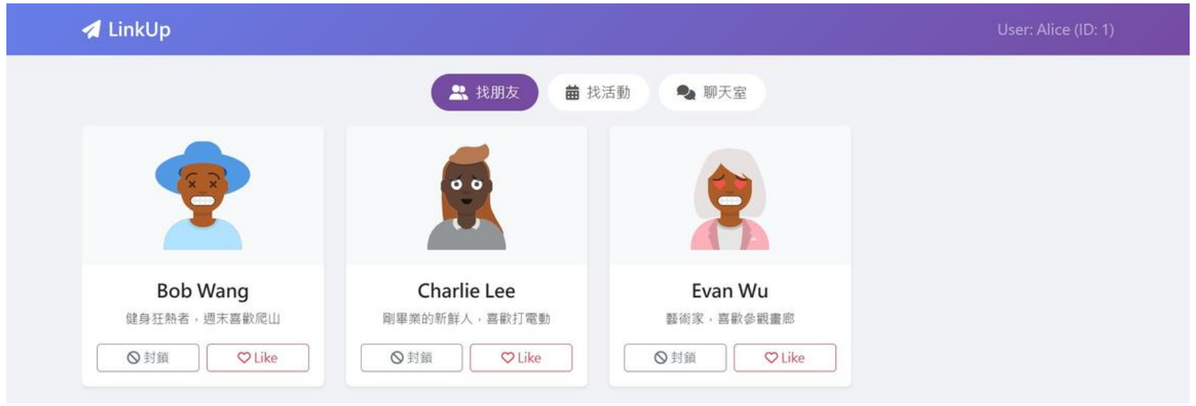


Block

	blocker_id	blocked_id	created_at
Edit Copy Delete	1	4	2026-01-05 15:51:39

(Block Table)

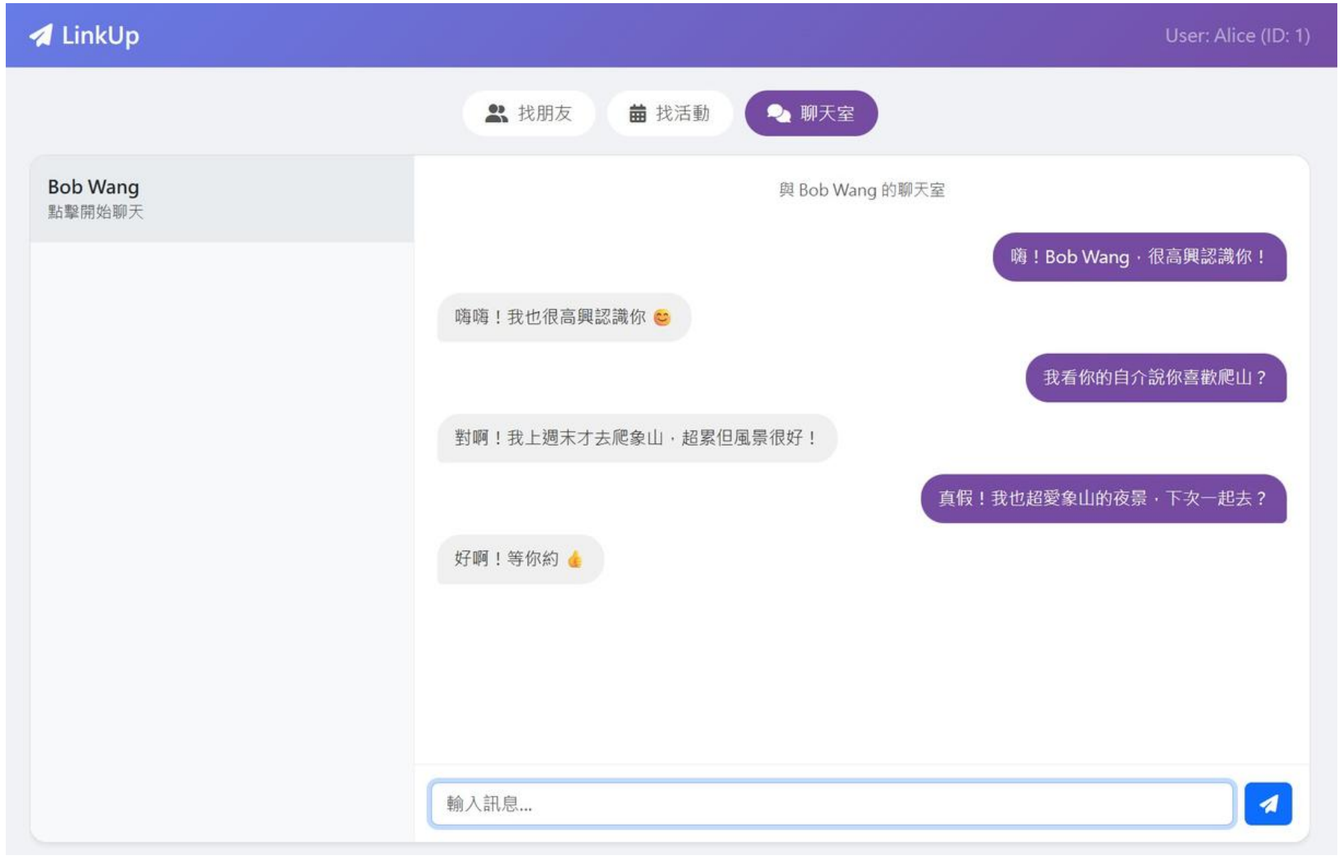




Chat

←T→	message_id	room_id	sender_id	content	sent_at
Edit Copy Delete	1	1	1	嗨！Bob Wang，很高興認識你！	2026-01-05 16:06:25
Edit Copy Delete	2	1	2	嗨嗨！我也很高興認識你 😊	2026-01-05 16:07:25
Edit Copy Delete	3	1	1	我看你的自介說你喜歡爬山？	2026-01-05 16:08:25
Edit Copy Delete	4	1	2	對啊！我上週末才去爬象山，超累但風景很好！	2026-01-05 16:09:25
Edit Copy Delete	5	1	1	真假！我也超愛象山的夜景，下次一起去？	2026-01-05 16:10:25
Edit Copy Delete	6	1	2	好啊！等你約 👍	2026-01-05 16:11:25

(Message Table)



Join an Event







	event_id	creator_id	title	description	event_date	max_participants
Edit Copy Delete	1	1	週末外拍團	去大安森林公園拍照，歡迎新手！	2026-06-01 14:00:00	5
Edit Copy Delete	2	2	象山夜景團	爬象山看101，結束後去吃消夜。	2026-06-05 19:00:00	10
Edit Copy Delete	3	1	期末舒壓聚餐	考完試了！一起去吃燒肉吧！	2026-06-10 18:30:00	4

(Event Table)

	event_id	user_id	joined_at
Edit Copy Delete	2	1	2026-01-05 16:19:57

(Event_Participant Table)

Others

Future Work

- 完善註冊與登入機制
- 完善活動發起功能
- 增加活動討論或交流區
- 加入檢舉與審查機制

THANK YOU

Group 3－LinkUp：交友配對與活動揪團平台